

Hidden Markov model approach for identifying the modular framework of the protein backbone

A.C.Camproux^{1,2,3}, P.Tuffery¹, J.P.Chevrolat²,
J.F.Boisvieux² and S.Hazout¹

¹Équipe de Bioinformatique Moléculaire, INSERM U155, Université Paris VII, case 7113, 2 Place Jussieu, 75251 Paris Cedex 05 and ²Département de Biomathématiques, CHU Pitié-Salpêtrière, 91 boulevard de l'Hôpital, 75013 Paris Cedex 13, France

³To whom correspondence should be addressed

The hidden Markov model (HMM) was used to identify recurrent short 3D structural building blocks (SBBs) describing protein backbones, independently of any *a priori* knowledge. Polypeptide chains are decomposed into a series of short segments defined by their inter- α -carbon distances. Basically, the model takes into account the sequentiality of the observed segments and assumes that each one corresponds to one of several possible SBBs. Fitting the model to a database of non-redundant proteins allowed us to decode proteins in terms of 12 distinct SBBs with different roles in protein structure. Some SBBs correspond to classical regular secondary structures. Others correspond to a significant subdivision of their bounding regions previously considered to be a single pattern. The major contribution of the HMM is that this model implicitly takes into account the sequential connections between SBBs and thus describes the most probable pathways by which the blocks are connected to form the framework of the protein structures. Validation of the SBBs code was performed by extracting SBB series repeated in recoding proteins and examining their structural similarities. Preliminary results on the sequence specificity of SBBs suggest promising perspectives for the prediction of SBBs or series of SBBs from the protein sequences.

Keywords: Markov chain/pattern classification/protein backbone/protein conformation/protein structure

Introduction

Protein conformation has long been the subject of experimental and computational interest. Looking at the database of known protein structures makes it clear that proteins use recurrent structural motifs at all levels of organization: secondary structure elements, local three-dimensional (3D) structures and protein domain topology (Unger *et al.*, 1989; Orengo *et al.*, 1993).

Standard secondary structure elements such as helices, strands and turns account for only 50–55% of all protein structures on average (Leszczynski and Rose, 1986). Attempts to categorize the remaining non-regular structures, called random coils, have led to the classification of several types of loops (Efimov, 1984; Fetrow, 1995). However, definition problems remain and the number of meaningful classes of elements, for example, varies between different studies. For predictive purposes, nonetheless, these are usually merged into just three categories: helices, strands and coils. However,

using the classical secondary structure classification scheme, fragments of a single class can vary substantially in their 3D structures (Unger *et al.*, 1989). Unger and Sussman (1993) have pointed out that a classification into 3D building blocks crosses the lines of traditional secondary structure assignments. Building blocks, unlike secondary structure elements, have tertiary meaning, because concatenating them in an overlapping manner produces a 3D chain. In addition, these authors suggest that building blocks may be deduced from the amino acid sequence more easily than secondary structure elements. An objective categorization of these structural blocks may lead to a deeper understanding of the modular architecture of proteins.

Automatic template-based classification procedures have identified a number of characteristic structural elements (Levitt and Chothia, 1976; Kabsch and Sander, 1983; Richards and Kundrot, 1988; Prestrelski *et al.*, 1992; Hutchinson and Thornton, 1993; Zhu, 1995). In general, however, structural similarities or dissimilarities between these motifs are not evident *per se*. Several teams have attempted to produce objective classifications of protein structures based on clustering of the three-dimensional coordinates or dihedral angle differences of the residues (Rooman *et al.*, 1989, 1990, 1991; Unger *et al.*, 1989; Fechteler *et al.*, 1995; Pavone *et al.*, 1996; Bystroff and Baker, 1998). Although these algorithms have been successful in identifying the classical helix and strand structures and, in some cases, new structural motifs, several problems such as the choice of a similarity criterion and a user-defined cut-off to determine final membership (Kelley *et al.*, 1996) may limit their usefulness. Recently, Fetrow *et al.* (1997) used an autoassociative neural network (autoANN) to extract local 3D structures from proteins, while Schuchhardt *et al.* (1996) applied a self-organizing Kohonen network to the same task. All these methods did not, however, take into account the local dependence of successive structural building blocks along the polypeptide chains.

In this work, the hidden Markov model (HMM) was used to identify the recurrent short structural 3D building blocks (SBBs) taking into account the local dependence between them. HMM became a subject of study in the late 1960s and was first applied to the problem of speech recognition (Baum *et al.*, 1970). These models have been successfully applied in molecular biology, in particular to distinguish coding from non-coding regions in DNA (Churchill, 1989), to generate multiple alignments for protein families and domains (Krogh *et al.*, 1994; Sonnhammer *et al.*, 1998), to predict secondary structures of proteins from amino acid sequences (Asai *et al.*, 1993; Stultz *et al.*, 1993; Karplus *et al.*, 1997) and to identify protein folds (Di Francesco *et al.*, 1997).

The aim of this study was to assess how much HMM is able to yield insight into the modular framework of proteins, i.e. to encode protein backbones into sequentially dependent structural blocks. We first used HMM to identify a reasonable

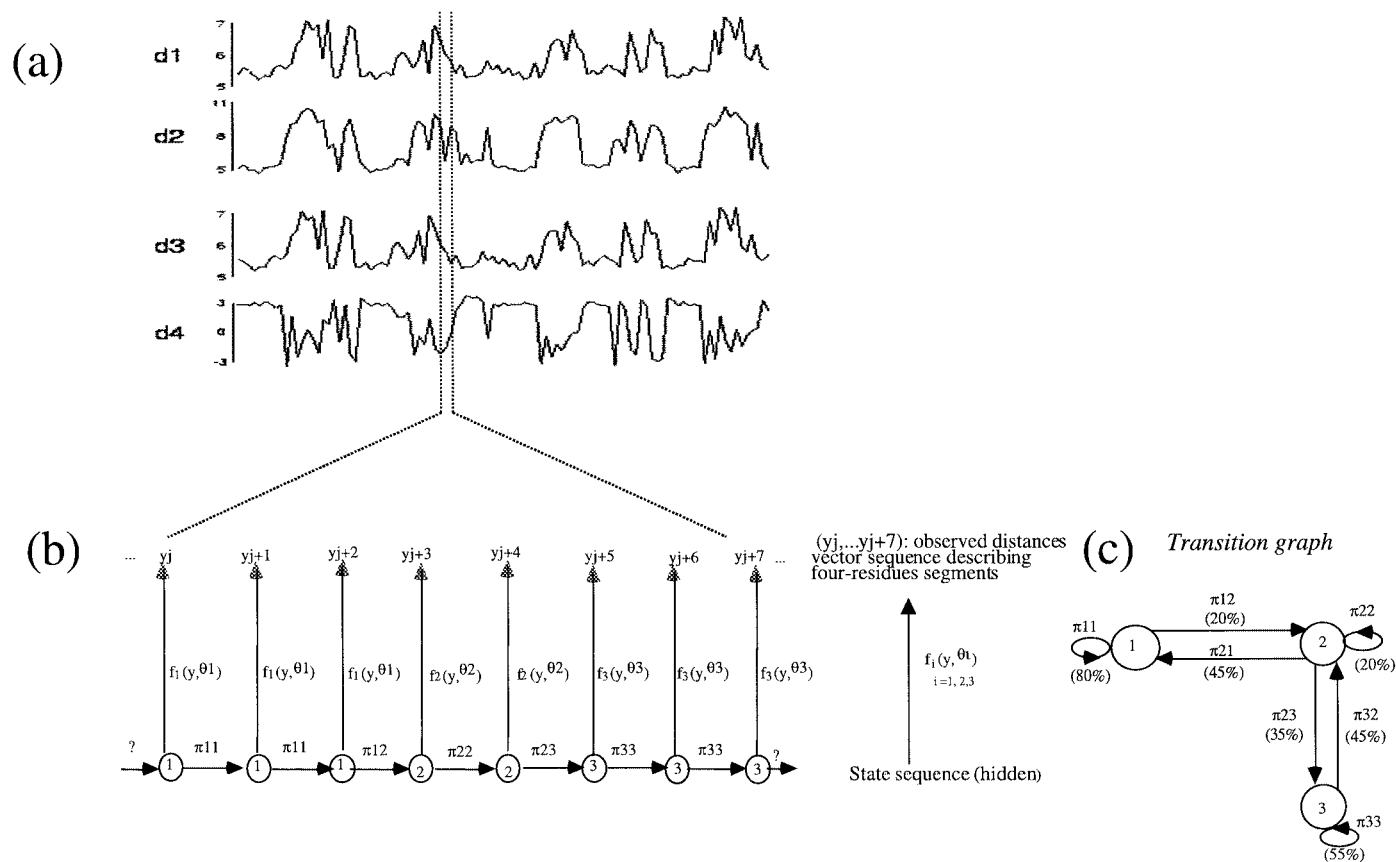


Fig. 1. (a) Example of four distance values observed (d_1, d_2, d_3, d_4) on a fragment of 100 four-residue segments of one protein (see comments in text). Some regions have similar profiles that suggest repeated structures. (b) A three-state hidden Markov model ($R = 3$) is illustrated on a fragment of eight four-residue segments ($y_j, y_{j+1}, \dots, y_{j+7}$) from a polypeptide chain. Each state $\{1, 2, 3\}$ emits four-dimensional distance vectors according to a multidimensional density probability $f_i(y, \theta_i)$, ($1 \leq i \leq 3$). This model infers a state sequence (1, 1, 1, 2, 2, 3, 3, 3) as a Markov chain corresponding to the series of distance vectors ($y_j, y_{j+1}, \dots, y_{j+7}$) describing the observed eight four-residue segments. (c) The estimated transition matrix $\pi = (\pi_{i,i'})_{1 \leq i, i' \leq R}$ of the Markov chain.

$$\begin{array}{c} 1 \quad 2 \quad 3 \\ \begin{array}{c} 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 0.80 & 0.20 & 0 \\ 0.45 & 0.20 & 0.35 \\ 0 & 0.45 & 0.55 \end{bmatrix} \end{array}$$

is schematized by arrows. State 1 is highly repetitive, with an 80% probability of repeating itself; state 3 is less repeated (55% of the time); state 2, however, is rarely repeated because it is essentially a transitory state. The transitions show no direct transition from state 1 to state 3

number of SBBs describing protein structures. We then studied how the identified SBBs are sequentially organized along the protein backbone. Finally, we checked if the sequences associated with SBBs are compatible with the amino acid preferences already observed for particular structures and tried to assess how much an SBB code could make sense in terms of the structural alphabet.

Materials and methods

Description of the database

Sequence data. A database of 100 non-homologous proteins was extracted from the Protein Data Bank (PDB) (Berstein *et al.*, 1977) to serve as a representative set for this study. The proteins selected met the criteria of a crystallographic resolution of <2.5 Å and a limited sequence homology ($\leq 25\%$ sequence identity with one another) (Hobohm *et al.*, 1992). Because the structure of the model is based on local dependence of successive residues in each protein, all non-contiguous protein chains, i.e. those containing segments that spanned gaps, were eliminated from the database. The PDB identifiers of the

proteins used are 1aaj, 1abk, 1add, 1ahc, 1arb, 1alkA, 1ast, 1atr, 1avhA, 1babB, 1bbpA, 1bet, 1bfg, 1bsaA, 1cdh, 1cmbA, 1colA, 1csh, 1cskA, 1cus, 1dhr, 1end, 1fdd, 1fkb, 1fxiA, 1gky, 1gmfA, 1gpr, 1hbq, 1hcrA, 1hlb, 1lisuA, 1l18, 1le4, 1lgaA, 1lis, 1loeB, 1mcta, 1mup, 1ndc, 1ofv, 1omp, 1onc, 1pfkA, 1pgb, 1pdg, 1poa, 1poc, 1poh, 1poxA, 1ppt, 1pspA, 1pyaB, 1rcb, 1rdh, 1ropA, 1sbp, 1sltA, 1syf, 1ten, 1tml, 1tndA, 1trkA, 1ttbA, 1vmoA, 2aaiB, 2acq, 2azaA, 2bopA, 2ccyA, 2cdv, 2chsA, 2cpl, 2cro, 2cyp, 2fal, 2gpb, 2hbg, 2hipA, 2ihl, 2liv, 2mge, 2mipA, 2mnr, 2pia, 2por, 2rslB, 2scpA, 2stv, 2ztaA, 3cla, 3gapA, 4fxn, 4gpb, 4insB, 5p21, 7pti, 8abp, 8rxnA, 9wgaA. The first four symbols represent the structural identifiers in the PDB database and the last symbol, when present, is the protein chain identifier. Information used in this paper is atomic coordinates, amino acid sequence and secondary structure assignments obtained from a prediction consensus (Colloc'h *et al.*, 1993).

Short protein backbone segments. The polypeptide chains were scanned in overlapping windows that encompassed four successive α -carbons C_α , thereby producing a succession of

short-backbone chain segments. As in some previous studies (Rooman *et al.*, 1990; Rackovsky, 1993; Pavone *et al.*, 1996; Smith *et al.*, 1997), we used four-residue lengths, which contain enough information to find basic structural elements: four-residue turns for α -helices, bridges for β -sheets and undefined loop structures. Moreover, a four-residue segment is small enough to keep the number of SBB categories reasonable. Increasing the number of residues per segment would introduce larger variability and would lead to a larger number of categories.

Structure descriptors of short protein segments. In this study, four distances $[d_1(j), d_2(j), d_3(j), d_4(j)]$ quantify the geometry for representing the j th, $1 \leq j \leq N$, four-residue segment that includes α -carbons $[C_{\alpha j+1}, C_{\alpha j+2}, C_{\alpha j+3}]$. $d_1(j), d_2(j), d_3(j)$ are distances between non-successive α -carbons, $(|U_{(j,j+2)}|, |U_{(j,j+3)}|, |U_{(j+1,j+3)}|)$, respectively, where $U_{(j,j')}$ denotes the vector defined by the α -carbon pair $(C_{\alpha j}, C_{\alpha j'})$, and $|U_{(j,j')}|$ denotes the modulus of this vector. The first and third distances d_1, d_3 are shifted from one residue and describe the opening of the beginning and end of the segments while d_2 describes the global length of the segments. The fourth distance d_4 is computed by a determinant of the three vectors defined by the successive C_α pairs normalized by the modulus of the first two vectors:

$$d_4(j) = \frac{\det(U_{(j,j+1)}, U_{(j+1,j+2)}, U_{(j+2,j+3)})}{|U_{(j,j+1)}| |U_{(j+1,j+2)}|}$$

This descriptor is proportional to the distance of the fourth C_α to a plane P built by the first three C_α ($C_{\alpha j}, C_{\alpha j+1}, C_{\alpha j+2}$) and provides information about the volume and the direction of the polypeptide fold. A 'flat' segment, i.e. with no volume, corresponds to a value of d_4 close to 0, contrary to a segment with clear volume. The sign of d_4 also indicates the topological orientation of the segment relative to P, trigonometric, i.e. the fourth C_α is located above P for a positive value of d_4 and inverse trigonometric, i.e. the fourth C_α is located below P, for a negative value of d_4 . These four descriptors $[d_1(j), d_2(j), d_3(j), d_4(j)]$, summarized in vector y_j , provide a unique representation of the conformation of each four-residue segment j .

Description of the hidden Markov model

Structure of the model. In this study, we use a stochastic model approach for classifying the 3D structures of four-residue protein segments and studying their relationship. The series $y_1, y_2, \dots, y_N$ of successive four-dimensional distance vectors describing the sequence of N four-residue segments obtained on the S polypeptide chains provide the basic data. Figure 1a shows the evolution of an observed signal ($y_j, y_{j+2}, \dots, y_{j+100}$) of 100 segments along a fragment of polypeptide chain. Similar regions appear in the successive profiles of the four observed signals, suggesting the existence of patterns specific to different local 3D structure.

Suppose we know that given polypeptide chains are made up of four-residue segments of R different types. We would then assume that there are R different states of the polypeptide chains, each state corresponding to a specific type of SBB. To take into account the existence of a 'local' dependence between contiguous SBBs, we assume a homogeneous Markov chain of order one to model a random sequence of states, which is the succession of underlying SBBs: the SBB at any position depends only on the SBB of the immediately preceding

position. In this study, we assume that a common Markov chain generates all proteins, meaning an identical dependence process between blocks, but we specify a starting process specific to each protein, i.e. each protein has its own initial block. Note, however, that this stochastic process is able to generate totally different blocks series, even starting from identical initial block. The evolution of the Markov chain is completely described by (1) the law $v_s = [v_s(i)]$ of the initial state of each polypeptide chain s , ($1 \leq s \leq S$), i.e. the probability that a polypeptide chain s starts in the R different states, and (2) the matrix of transition probabilities $\pi = (\pi_{ii'})$ $1 \leq i, i' \leq R$ between R different states of the Markov chain, where $\pi_{ii'} = P(X_j = i' | X_{j-1} = i)$ is the probability of different proteins evolving from state i to i' at any position j . The transition matrix of the Markov chain results in $R \times (R - 1)$ parameters. At the same time, because the law of initial state is assumed specific to the start process of each polypeptide chain s , ($1 \leq s \leq S$), it results in $(R - 1) \times S$ parameters.

Finally, each state i , ($1 \leq i \leq R$), of the Markov chain generates segments following a continuous multidimensional density $f_i(y, \theta_i)$ of unknown parameter θ_i , meaning that to each SBB of the proteins corresponds an average four-residue segment with specific variability. Four-dimensional Gaussian density was chosen as the continuous density $f_i(y, \theta_i)$ to ensure that parameters can be estimated in a consistent way (Rabiner, 1989). This results in unknown 14-dimensional parameter $\theta_i = (\mu_i, \Sigma_i)$, ($1 \leq i \leq R$). Indeed, the average four-residue segments conformation associated to SBB i is described by an average vector $\mu_i = (\mu_{i1}, \mu_{i2}, \mu_{i3}, \mu_{i4})$ of the four-distance values (d_1, d_2, d_3, d_4) between residues and variability which is quantified by a symmetric covariance matrix, $\Sigma_i = [\sigma_{i(\delta, \delta')}]$, $1 \leq \delta, \delta' \leq 4$.

In the example presented in Figure 1b, we focus on a fragment of eight segments ($y_j, y_{j+1}, \dots, y_{j+7}$) from a polypeptide chain and we assume they derive from only three SBBs ($R = 3$). In this example, the transition matrix of the hidden Markov model, illustrated in Figure 1c, shows no transitions between states 1 and 3 which are relayed only by state 2.

Finally, an HMM with R states obtained on S proteins yields $(R - 1) \times S + (R - 1 + 14) \times R$ parameters. For example, an HMM with 12 SBBs obtained on 100 polypeptide chains (S) yields 1400 parameters: $(R - 1) \times S + (R - 1 + 14) \times R = 1400$.

Estimation of model parameters. To compare the conformations associated with the blocks and the actual conformations of the fragments, we calculate the probability of observing the 3D structure of different fragments for a given model defined by parameters. The corresponding density of probability computed on all the observed fragments is the likelihood of the model. Hence maximizing the likelihood is equivalent to finding the optimum set of parameters of the model that approximates best the actual protein conformations. To determine the number of states R , we start with a simple model, for instance $R = 3$ and consider models in ascending order of complexity, by increasing R . Comparison of different models is based on likelihood, using Bayesian information criteria (BIC) (Schwartz, 1978). Unknown parameters v_s , ($1 \leq s \leq S$), π , and θ_i , ($1 \leq i \leq R$) of the HMM are estimated by maximizing the complete likelihood of the model, computed recursively by exploiting the underlying Markov structure of the model, following the algorithm of Baum *et al.* (1970). For an overview of the basic theory of HMM and practical details on methods of implementation of the theory, see Rabiner (1989).

Reconstructing proteins by SBBs. The resulting model is a hidden Markov model: the SBBs sequence has not been observed (it is hidden). Only the four-dimensional distance vector sequence $y_1, \dots, y_N$ that is generated by the hidden series of SBBs is observed. Our ultimate goal is to reconstruct the unobserved SBB sequences of the polypeptide chains, given the observed sequence $y_1, \dots, y_N$. The obvious approach is to select a model (i.e. to determine the number R of SBBs), to estimate its parameters and then to find the most probable path of SBBs among all possible paths in $\{1, \dots, R\}^N$ using the Viterbi algorithm (Rabiner, 1989; Camproux *et al.*, 1996).

Results

Description of the SBB categorization and of their organization

Number of SBBs. Fitting the HMM to the 19 017 experimental four-residue segments obtained from the database of 100 polypeptide chains resulted in a representative set of 12 distinct SBBs. As a first step, we used a number of states of three ($R = 3$). Progressively increasing the number of states up to 12 resulted in a significant improvement of the BIC. The simplest model was able to identify only one SBB corresponding to α -helices and two non really specific SBBs. When increasing R up to 12, HMM progressively identified β -strands and decomposed coil regions. Limiting the number of SBBs to 12 allows one to obtain representative SBBs corresponding to at least 3% of the segments database and keep a reasonable number of parameters: the corresponding HMM estimated on the 100 different protein chains database yields 1400 parameters (see Materials and methods). The generalizability of the results, concerning 12 identified SBBs, was checked by fitting the HMM to two non-overlapping datasets of 50 different polypeptide chains and by comparing the SBB code obtained. This identified very similar SBBs (only three blocks present more than one distance value significantly different on the two independent 50 protein samples) suggesting that 100 polypeptide chains are sufficient to identify the SBBs code.

Structural description of SBBs. Table I describes the structural characteristics of the average four-residue segment associated with the 12 SBBs identified by HMM together with their correspondence to the usual secondary structures (a segment is classified in one secondary structure when its third central residue carbon is assigned to it). For clarity's sake, the SBB nomenclature is roughly based on the correspondence to the usual secondary structure, they are labelled ($\alpha_1, \alpha_2, \alpha'_-, \alpha'_+, \alpha'_\gamma_1, \gamma_2, \gamma_{\beta\alpha}, \gamma_\beta, \gamma_{\alpha\beta}, \beta_2, \beta_1$). SBBs were presented relatively to the length of their associated segments which globally increases from SBB α_1 to β_1 . This table also reports a similarity index within each SBB (r.m.s.d.w.) and a dispersion index between different SBB pairs (r.m.s.d.b.).

The 12 SBBs appear well identified: a multivariate analysis computed on the four distances confirms that segments associated with SBBs are significantly different, globally and pairwise (multivariate variance analysis and Hotelling T -test with $p < 0.001$). The standard deviations of the distances are low: <0.50 Å for the majority. The r.m.s.d.w. varies in the range 0.09 – 0.84 Å for all SBBs, below the standard threshold of 1 Å (Unger *et al.*, 1989; Unger and Sussman, 1993). Moreover, the segments associated with the different SBBs appear relatively distinct in terms of structure but close within each block: all the r.m.s.d.b. between SBB pairs are larger than the two r.m.s.d.w. of the SBB pairs, except for SBB pairs (α_1, α_2), (γ_β, γ_2), (γ_β, β_1). This point is further discussed in the section

‘Analysing the SBB connections’ since HMM performs SBB classification taking into account the sequentiality of the SBBs and not only geometrical aspect.

SBBs α_1 and α_2 , representing 37.9% of the observed four-residue segments, provide the segments for 91.6% of the α -helices. Their associated segments have the lowest means and variabilities for the first three distances and the highest mean for the volumetric parameter, implying a short, bulky and regular structure with a clear trigonometric orientation, in agreement with their α -helices correspondence. Segments associated with SBB α_1 have the smallest variability (r.m.s.d.w. = 0.09 Å) while SBB α_2 appears to correspond to somewhat more irregular and less compact structure (r.m.s.d.w. = 0.2 Å).

SBBs β_1 and β_2 contain 23.8% of the segments. They appear mostly located in β -strands: they provide the segments for 78.7% of the β -strands. They correspond to well-defined stretched structures (r.m.s.d.w. ≤ 0.35 Å). The geometry of β_1 (no volume) is compatible with flat β -strands while SBB β_2 describes a somewhat shorter structure with a clearly negative trigonometric orientation.

To simplify descriptions, SBBs α_1 and α_2 , clearly associated with α -helices are artificially gathered in cluster A and SBBs β_1 and β_2 , associated with β -strands, gathered in cluster B.

SBBs α'_-, α'_+ and α'_γ each represent about 3% of the observed segments. Their associated segments represent about 7% of the helices and 16% of the coils. Corresponding segments are geometrically close to those of cluster A but suggest longer forms. Segments associated with SBBs α'_+ and α'_γ present larger length and variability than SBBs α'_- and are distinguishable by their opposite topological orientation.

Last, transitory SBBs γ appear located in more than 80% in coils (data not shown). Especially SBB γ_1 appears almost exclusively in coils. SBB $\gamma_{\alpha\beta}$ is noteworthy because of its high occurrence frequency (11.4%), whereas others represent about 4% each of the observed segments, and because it takes into account about 20% of the coils. The r.m.s.d.w. values of these blocks are acceptable (0.33 – 0.84 Å). Only $\gamma_{\beta\alpha}$ and $\gamma_{\alpha\beta}$ describe segments with volume and clear trigonometric orientation, negative for $\gamma_{\beta\alpha}$ and positive for $\gamma_{\alpha\beta}$.

Segments associated with SBBs γ_1, γ_2 and $\gamma_{\alpha\beta}$ present small d_3 values close to cluster A and longer d_1 values close to cluster B. They are distinguishable by different d_2 values. Segments associated with SBBs γ_β and $\gamma_{\alpha\beta}$ have d_3 values close to cluster B but are distinguishable with other parameters, especially d_1 close to cluster B or A. SBB $\gamma_{\alpha\beta}$ describes a structure inverse to $\gamma_{\beta\alpha}$ (inverted d_1 and d_3 values and opposite orientations); SBBs γ_β and $\gamma_{\alpha\beta}$ correspond to the less well defined structures ($0.84, 0.73$ Å). In particular, their d_3 values present large variabilities ($9.13 \pm 0.89, 8.25 \pm 0.94$ Å).

Analysing the SBB connections. Table II reports estimated transitions between the identified SBBs and average number of times each SBB i is repeated, computed by $[1/(1-\pi_{ii})]$ with π_{ii} the probability of staying in SBB i . An estimated average number of repetitions equal to 1 implies a transitory block (no repetition of the structure).

As a first observation, the estimated transition matrix is relatively sparse; $<30\%$ of the estimated probability transitions occur relatively often ($>10\%$). This results in a limited number of possible connections. In particular, clusters A and B describing regular secondary structures correspond to the most repeated structures while other blocks appear rather like

Table I. Description of the four-residue segments associated with the 12 SBBs identified with (1) the mean and standard deviation of the four distance values of the average conformational four-residue segment in Å; (2) a similarity index within each SBBs (r.m.s.d.w.), as estimated from the average r.m.s.d. obtained on a sample of its associated segments; (3) a dispersion index between different SBBs, as estimated from the minimum value of the r.m.s.d. between different pairs of SBBs (r.m.s.d.b.); (4) the proportion and number of corresponding four-residue segments; and (5) correspondence between SBBs segments and the usual secondary structures

SBBs Corresponding four-residue segments

	Mean $\pm$ standard deviation (Å) of the four distances describing the average conformational four-residue segments				R.m.s.d.w. (Å)	R.m.s.d.b. (Å)	Frequency (19017) % (N)	Percentage of third residues assigned to a given secondary structure		
	d_1	d_2	d_3	d_4				α (32.9)	Coil (47.2)	β (19.9)
α_1	5.45 $\pm$ 0.11	5.13 $\pm$ 0.16	5.45 $\pm$ 0.11	2.92 $\pm$ 0.17	0.09	0.15	23.07 (4387)	63.3	4.5	
α_2	5.48 $\pm$ 0.21	5.43 $\pm$ 0.35	5.53 $\pm$ 0.21	3.00 $\pm$ 0.40	0.20	0.15	14.84 (2823)	28.3	11.6	0.8
α'	5.81 $\pm$ 0.33	5.59 $\pm$ 0.47	5.91 $\pm$ 0.28	1.66 $\pm$ 0.60	0.26	1.4	3.52 (669)	3.3	5	0.2
α'_-	5.57 $\pm$ 0.27	7.40 $\pm$ 0.98	5.65 $\pm$ 0.26	-3.18 $\pm$ 0.48	0.56	0.87	2.98 (567)	0.5	6	0.3
α'_+	5.64 $\pm$ 0.30	7.46 $\pm$ 0.83	5.67 $\pm$ 0.38	3.38 $\pm$ 0.44	0.38	1.11	3.63 (691)	3.4	5.2	0.2
γ_1	6.65 $\pm$ 0.38	6.71 $\pm$ 1.18	5.61 $\pm$ 0.27	-0.24 $\pm$ 1.71	0.59	1.2	2.98 (567)	0.5	5.9	0.2
γ_2	6.20 $\pm$ 0.41	9.10 $\pm$ 0.32	5.67 $\pm$ 0.26	-0.18 $\pm$ 0.97	0.38	0.64	4.29 (816)		7.6	3.1
$\gamma_{\beta\alpha}$	6.68 $\pm$ 0.31	8.57 $\pm$ 0.47	5.55 $\pm$ 0.26	-2.54 $\pm$ 0.53	0.33	0.63	5.62 (1068)		10.5	2.4
$\gamma_{\alpha\beta}$	5.69 $\pm$ 0.27	8.25 $\pm$ 0.94	6.74 $\pm$ 0.32	1.60 $\pm$ 1.54	0.73	1.11	11.44 (2175)	0.7	19.5	10.2
γ_β	6.81 $\pm$ 0.32	9.13 $\pm$ 0.89	6.71 $\pm$ 0.40	-0.61 $\pm$ 1.68	0.84	0.47	3.81 (724)		6.7	3.8
β_2	6.74 $\pm$ 0.31	9.40 $\pm$ 0.47	6.46 $\pm$ 0.26	-2.36 $\pm$ 0.48	0.32	0.93	8.77 (1667)		9.8	21.1
β_1	6.65 $\pm$ 0.31	10.11 $\pm$ 0.34	6.74 $\pm$ 0.30	-0.65 $\pm$ 0.67	0.34	0.47	15.05 (2862)		7.6	57.6

A four-residue segment is classified in one secondary structure when its third central residue carbon is assigned to it. The proportions of segments assigned in the database to α -helices, β -strands and coils are 32.9, 19.9 and 47.2, respectively.

Table II. Transition matrix: estimates of transition probabilities between the 12 SBBs together with the average number of repetitions of each SBB

SSBBs	α_1	α_2	α'	α'_-	α'_+	γ_1	γ_2	$\gamma_{\beta\alpha}$	$\gamma_{\alpha\beta}$	γ_β	β_2	β_1	Average number of repetitions
α_1	83.9	12.2		2.9	0.2				0.7				6.3
α_2	17.0	43.0	11.0	5.0	10.0				14.0				1.7
α'	0.3	7.3	19.5	12.4	13.9		8.9	0.7	37.0				1.2
α'_-		0.8	1.5	5.4	16.0	2.3	5.0		69.0				1
α'_+	4.5	37.0	10	4	9.7		12.2		22.6				1.1
γ_1	2.5	13.7	7.7	14.5	15		7.3		39.3				1
γ_2	2.8	33.9		5.0	8.4		6.0		43.9				1.1
$\gamma_{\beta\alpha}$	12.1	44.3		2.0	3.4		6.7		31.5				1
$\gamma_{\alpha\beta}$				0.2	0.2	9.0	6.6	16.8			29.5	25.6	1
γ_β			2.0		0.2	22.0	4.0	11.0		30.9	17.9	12.0	1.4
β_2			4.5			4.1	9.6	23.6			21.0	37.2	1.3
β_1						4.9	5.0	8.0		9.1	18.3	54.7	2.2

transitory structures. No direct transitions are observed and there are only a few possible pathways between them.

Globally, cluster A exits towards to $\gamma_{\alpha\beta}$ either directly or indirectly after some paths in one of the three SBBs α' , α'_- and α'_+ which present few switchings. From SBB $\gamma_{\alpha\beta}$, proteins transit directly to cluster B or enter a path through other SBBs γ . From cluster B, proteins transit to SBBs $\gamma_{\beta\alpha}$, γ_1 or γ_2 which are the preferred ways of entrance in cluster A. There are some paths within SBBs γ , associated with coil regions, mainly unidirectional. A simplified representation of the main paths

between SBBs, in terms of occurrences within the database, is presented Figure 2 and illustrates the asymmetry of the connections.

Concerning cluster A, the inter-connection of SBBs α_1 and α_2 is strong. Entries to this cluster are mainly performed through α_2 (from SBBs $\gamma_{\beta\alpha}$, α'_+ , γ_2 and γ_1) or by a transition from $\gamma_{\beta\alpha}$ to α_1 . Exits are mostly performed through α_2 (to α' , α'_+ or $\gamma_{\alpha\beta}$) or by direct transition from α_1 to α'_- . Interpreting this together with geometrical characteristics in terms of α -helices suggests that repeated α_1 (average number of 6.3

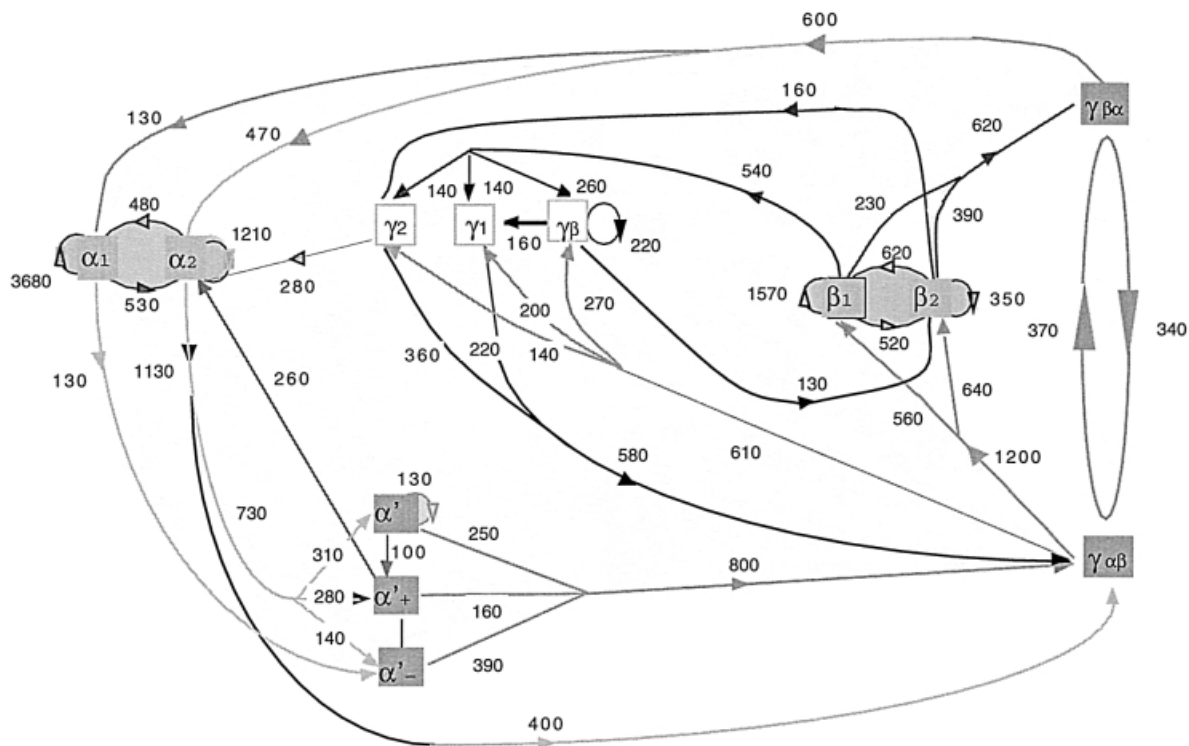


Fig. 2. Simplified network of main paths between SBBs. Only the paths which occur more than 100 times within the set of proteins are illustrated.

repetitions) corresponds to the core of α -helices while α_2 is the preferred blocks through which connections are made to other blocks. In fact, a more precise analysis of the paths from SBBs $\gamma_{\beta\alpha}$ or γ_1 and γ_2 and α'_+ into α_2 shows it is then either involved in α -helices by leading to α_1 (about 30%) or in paths within coils (about 70%). When α_1 leads to α_2 , there is only 11% of returns into α_1 . This suggests that block α_2 may correspond either to extremities of α -helices (but not yet to bounding regions) or to slight deformation within α -helices or to helical form within coils. The average length of an α -helix in this database, 11.6 ± 5.6 residues, is compatible with the estimated average number of residues corresponding to series of SBB α including one or two SBBs α_2 of 11 or 12.7 residues, respectively (for instance 6.3 repetitions of SBBs α_1 plus 1.7 for α_2 plus 3 since SBBs correspond to a four-residue length).

Concerning cluster B, SBBs β_1 and β_2 both participate in β -strands. They are strongly inter-connected and have both ways of connections to other blocks. Their inputs come from $\gamma_{\alpha\beta}$ and γ_{β} . Concerning outputs, if both blocks have possible transitions towards $\gamma_{\beta\alpha}$, they also have specific exits: towards γ_{β} for SBB β_1 and towards γ_1 for β_2 . Proteins stay in SBB β_1 in about 55% of the cases, for an average number of 2.2 repetitions; SBB β_2 is repeated less (average of 1.3 repetitions). Considering β -strands are decomposed into both SBBs β_1 and β_2 , the additive average number of repetitions is about 3.5 (~ 2.2 plus 1.3), corresponding to an average number of residues of 6.5, in agreement with the observed length of β -strands in the database: 6.4 ± 2.8 residues.

SBBs $\alpha'_$, α'_+ and α'_- constitute transitory blocks, mainly located on exit from cluster A. They present some switching and lead mainly into $\gamma_{\alpha\beta}$. This suggests that they may correspond to different α -helix C-terminating structures: SBB α'_- is rather on exit from α_1 and owing to its negative trigonometric orientation it should correspond to an abrupt breaker of α -helices. SBB $\alpha'_$ presents a structure more similar to segments

from cluster A and can lead to α'_+ and α'_- ; this suggests that it may correspond to a more progressive termination helicoidal structure. SBB α'_+ is found on both entry and exit of SBB α_2 .

SBBs γ , describing coil regions, constitute mainly transitory blocks with unidirectional connections. We can note some trends of these blocks relative to regular secondary structures, in agreement with their geometrical characteristics (see Table I). SBBs γ_2 , γ_1 and $\gamma_{\beta\alpha}$ are rather involved in connections from α -helices to β -strands while SBBs $\gamma_{\alpha\beta}$ and γ_{β} are rather involved in connections from β -strands to α -helices. Other inputs in blocks $\gamma_{\beta\alpha}$, γ_1 and γ_2 come from SBBs γ_{β} and $\gamma_{\alpha\beta}$ while other outputs lead to SBB $\gamma_{\alpha\beta}$. Some inputs in the C-cap α -helices blocks come from SBB γ_1 and some outputs lead to SBB γ_2 . Inputs clearly differ for the two blocks $\gamma_{\alpha\beta}$ and γ_{β} : the only inputs in SBB γ_{β} are $\gamma_{\alpha\beta}$ and β_1 . Transitory block $\gamma_{\alpha\beta}$ appears to be the terminal exit from α -helices (directly or indirectly after some paths in C-cap α -helices blocks) and has also a large number of connections with SBBs (γ_2 , γ_1 and $\gamma_{\beta\alpha}$), particularly as input. Interestingly, switches occur between these two sets of blocks but few within, confirming that SBBs describing coil regions are really specific. This suggests strongly that HMM has performed more than splitting the conformational space and is able to detect also some organization.

Do SBBs exhibit some sequence specificity? Figure 3 illustrates the sequence-specificity matrices. Each matrix consists, for each SBB, in the distribution of amino acids at each of the four positions, normalized by the distribution of the amino acids in the whole database. No matrix is random, i.e. building blocks reveal high or low frequencies of particular amino acids in various positions of their four-residue segments. Some SBBs associated with coils show only weak specificity. Nonetheless, SBBs are distinguishable and characterizable in terms of

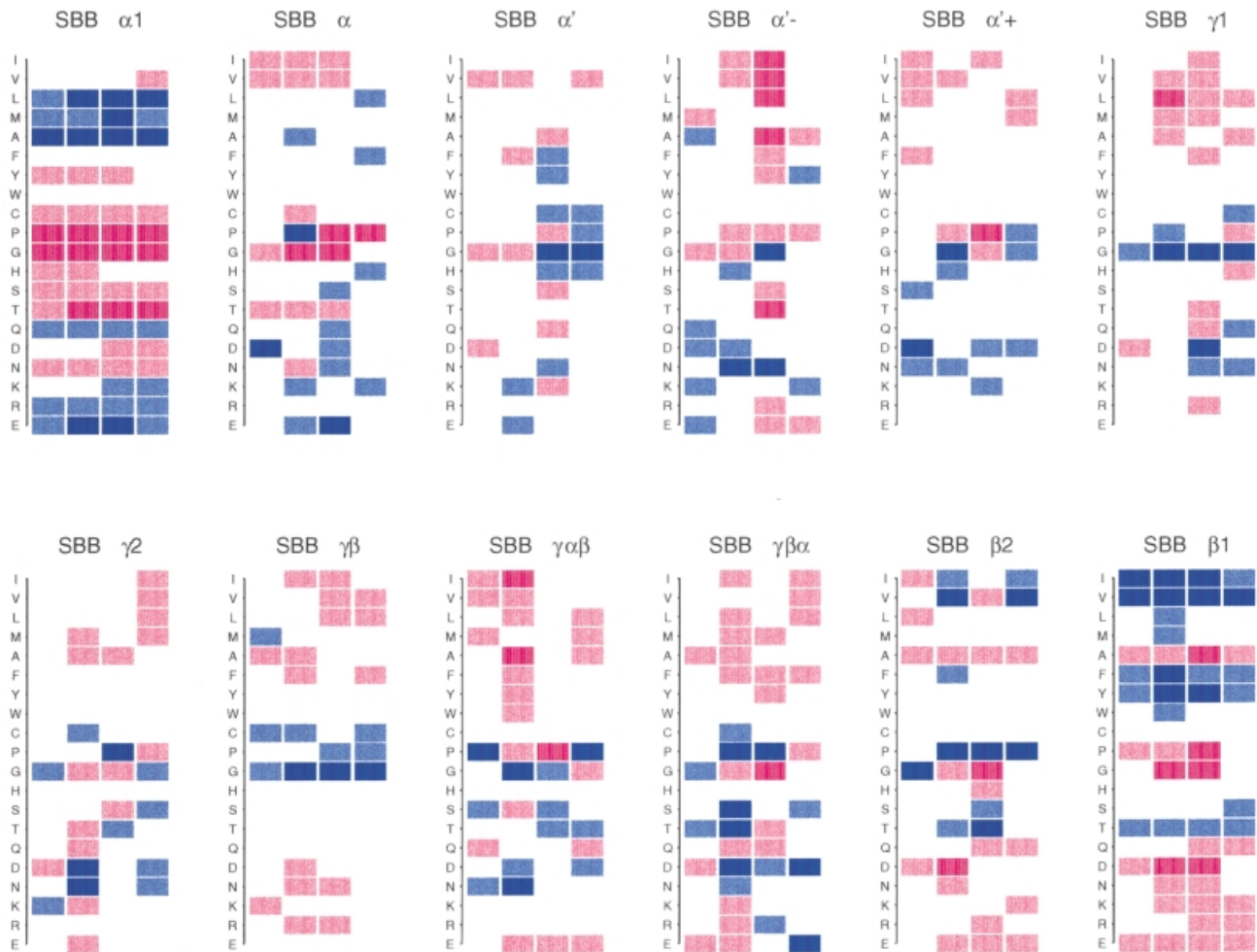


Fig. 3. Frequency of amino acid occurrence at each of the four positions of the segment associated with each SBB normalized into Z-scores relative to frequency observed in the database. The amino acids, represented by their one-letter codes, are plotted on the y-axis and the positions within each segment (1–4) on the x-axis. The colours of each square indicate the levels of occurrence of each amino acid at each position. Higher levels are indicated by dark colours (absolute Z-scores > 4) then pale shades ($2 \leq$ absolute Z-scores ≤ 4) and low levels by white (absolute Z-scores < 2). Blue squares designate over-represented amino acids (Z-scores > 2) and pink squares under-represented amino acids (Z-scores < -2).

sequence specificity. Clusters A and B contain the most specific sequence patterns. We also note that repeated SBBs α_1 and β_1 , tend to have uniform frequencies of occurrence for particular amino acids along of the four successive positions.

Are the specificities in agreement with well-known patterns? Analysis of the amino acids favoured or disfavoured in clusters A and B, associated with regular secondary structures, appear globally consistent with the known amino acid preferences in terms of secondary structures (Argos and Palau, 1982; Richardson and Richardson, 1988; Presta, 1989; Aurora *et al.*, 1994; Seales *et al.*, 1994).

SBB α_1 strongly prefers the hydrophobic (alanine, leucine, methionine) and charged (glutamic acid and arginine) amino acids, as well as glutamine. Proline and glycine residues, known as α -helix breakers, are strongly disfavoured in all positions as is threonine. Cysteine, serine, asparagine and threonine are also disfavoured. SBB α_2 exhibits less clear specificity; this may be related to its different role (see connection matrix section). This block presents contrasted sequence specificity, mostly by the successive overrepresentations of aspartic acid, proline and glutamic acid and underrepresentations of hydrophobic residues, particularly of isoleucine and valine. This is in agreement with one isolated α -turn structure described by Pavone *et al.* (1996).

Alanine and charged residues are disfavoured in both β_1 and β_2 . Isoleucine, valine, threonine, phenylalanine and tyrosine appear to be overrepresented in β_1 . Threonine, consistently with the observations of Argos and Palau (1982), appears to be distributed uniformly. Phenylalanine and tyrosine show a central preference, as reported by Unger *et al.* (1989). We observe that the hydrophobic non-polar acids favoured in SBB β_1 are disfavoured in SBB α_1 and vice versa. SBB β_2 , which corresponds to less regular β -strands, is more contrasted than SBB β_1 and presents a clearly different specificity: this block strongly favours glycine then proline together with threonine, valine and isoleucine.

Sequence distribution matrices of other SBBs have been obtained on smaller samples of segments (see Table I) and thus their sequence specificity is less informative. However, they do exhibit different sequence specificity. Globally they tend to disfavour hydrophobic acids favoured by regular secondary structures.

Concerning SBBs α' , α_- , α'_+ , their structural features and connections suggest that they are located on exit of α -helices. Glycine appears strongly favoured at some locations, often preceded by a histidine. This can be related to previous observations that the C-cap position is overwhelmingly dominated by glycine (Richardson and Richardson, 1988; Aurora

et al., 1994). Concerning different sequence specificity, SBB α'_- shows a preference for alanine, lysine and glutamic acid at the first position, which could be related to one C-cap structure identified by Fetrow *et al.* (1997). It also has a strong under-representation of hydrophobic amino acids at location 3. Consistent with the fact that α' may correspond to progressive termination helicoidal structures, this block favours glycine, proline and cysteine only at the last positions. For α'_+ , which sometimes corresponds to entrance in helices, proline is also over-represented in the last position.

Concerning SBBs γ_1 , γ_2 and $\gamma_{\beta\alpha}$, they were identified as putative N-cap α -helix structures. They strongly favour proline at some positions, which is consistent with Richardson and Richardson (1988). Other preferences are mainly for aspartic acid and asparagine, consistent with their possible role of β -strand C-cap (Fetrow *et al.*, 1997). Also, SBB $\gamma_{\beta\alpha}$ exhibits a strong preference for serine, threonine and glutamic acid, as previously reported (Harper and Rose, 1993; Seales *et al.*, 1994; Fetrow *et al.*, 1997). SBB γ_1 exhibits a strong preference for glycine.

SBBs $\gamma_{\alpha\beta}$ and γ_{β} , most often found as termination of the β -strands, reveal preferences for proline, glycine (alternately over- and under-represented for complementary positions of $\gamma_{\beta\alpha}$), asparagine, serine and threonine. As described by Fetrow *et al.* (1997) for its identified strand N-cap structure, SBB $\gamma_{\alpha\beta}$ exhibits a preference for charged residues at some locations. SBB γ_{β} , not found as exit of helices, shows a strong preference for proline, glycine and cysteine residues.

Can SBBs code form a structural alphabet? The HMM allows one to find the most likely path of SBBs within each protein, taking into account the SBB sequentiality (see Materials and methods). All four-residue segments of the database are recoded in terms of these 12 blocks, so that each protein chain can be described as a series of SBBs. For illustrative purposes, the structure of an L-arabinose binding protein (8abp), which is an $\alpha\beta$ protein, is displayed in Figure 4, decoded in terms of SBBs. It is coloured on the left side according to its assignment by this model, that is, the SBB category, and on the right side according to its conventional secondary structure.

Does the 'SBB code' obtained by the model make sense in terms of extraction of 3D structural similarity through identification of series of SBBs, long structural block (LSB), repeated in different proteins? Analysis of the SBB preference within four types of connection fragments located between regular secondary structures, helices–helices, helices–strands, strands–strands and strands–helices, were performed together with studies on short SBB series specific to these different types of connection fragments. This work confirmed that a 12 SBB code can yield information on the structures of the coils flanking regular secondary structures (Camproux *et al.*, 1999). More general information can come from a search for contiguous backbone chain fragments of fixed length defined by identical SBB words in the set of proteins and then assessing their 3D superimposition. The extractions are identified by two parameters: the fragment length L , i.e. the number of SBBs in each series, and the number of occurrences Nb of this series.

We present the results for some LSBs with $L \geq 11$ (including more than 14 residues) and $Nb \geq 2$, using an exact match between words. Hence the structural diversity of the isolated patterns only comes from the variability within blocks. Table III describes 12 LSBs extracted from the database, reporting

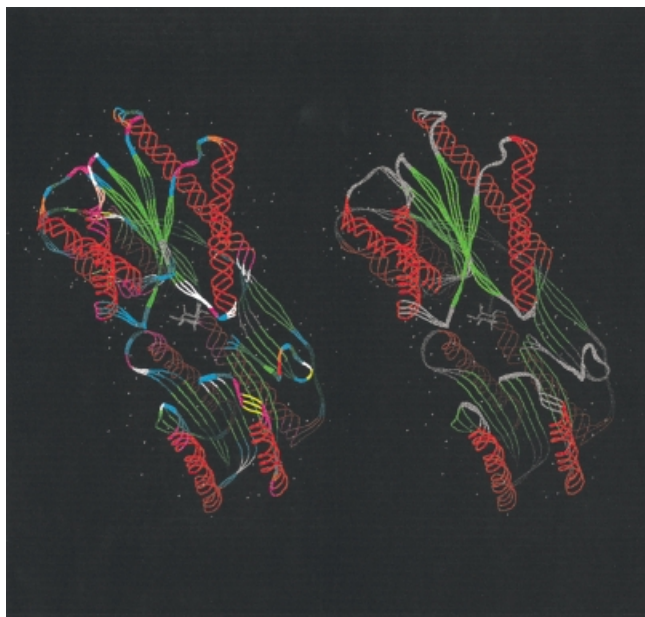


Fig. 4. Illustration of the structure of an L-arabinose binding protein (8abp), an $\alpha\beta$ protein coloured according to its assignment by the estimated hidden Markov model, that is, the SBB category of the third residue in each four-residue segment (left side): α_1 , red; α_2 , pink; α' , pale pink; α'_- , orange; α'_+ , yellow; γ_1 , brown; γ_2 , white; γ_{β} , grey; $\gamma_{\alpha\beta}$, sky blue; $\gamma_{\beta\alpha}$, blue/green; β_2 , green; β_1 , light green. On the right side, residues are coloured according to their conventional secondary structure assignment: helix, red; strand, green; coil, grey. α -Helices are mostly classified as SBB α_1 (red) except for those on the extremities, which are mainly assigned to SBB α_2 (pink). There are also a few direct transitions from SBB α_1 (red) towards SBB α'_- (orange). The β -strands are often preceded by SBB $\gamma_{\alpha\beta}$ (sky blue). As expected, cluster B divides β -strands into more or less stretched and regular β -strands. Indeed, some β -strands are completely labelled as SBB β_1 (light green) or SBB β_2 (green) and once as SBB γ_{β} (grey). The figure was generated using Xmmol (Tufféry, 1995).

their lengths (L), their number of occurrence (Nb) in the database and their associated average pairwise $\langle \text{r.m.s.d.} \rangle$ deviations, denoted r.m.s.d. The right-hand side of the table provides for each LSB (1) its protein PDB code, followed by the position of the first residue in the PDB entry and (2) the full amino acid sequence corresponding to each repetition with the conserved sequence pattern indicated. Analysis of the amino acid sequences associated with each LSB sometimes reveals patterns of strictly conserved amino acid residues. Moreover, the amino acid sequences display some positions preferentially occupied by charged, hydrophobic or glycine residues. Other conserved features involve aliphatic, aromatic or small residues. All the LSBs had fairly low 3D structural variability ($\text{r.m.s.d.} \leq 1.1 \text{ \AA}$ apart from LSB 4) and some extracted patterns (LSBs 2, 8, 10 and 12) appear extremely homogeneous, with $\text{r.m.s.d.} < 0.6 \text{ \AA}$. Figure 5 presents these patterns coloured according to their corresponding secondary structures. These 12 LSBs correspond to recurrent patterns in regions bounding regular secondary structures as well as in random coils regions. This figure shows how similar the different members of these LSBs are and illustrates how the SBB code allows one to extract similar patterns.

Discussion

Comparison of SBBs with other structural classifications

Most other studies (e.g. Richards and Kundrot, 1988; Unger *et al.*, 1989; Prestrelski *et al.*, 1992; Unger and Sussman,

Table III. Description of extracted repeated long structural building blocks

LSB	L	<r.m.s>d	name	pos	S E Q U E N C E S
(Nb)					
1	16	0.77	2mipA	9	P V V T A Y I E G Q P V E V L L
	(2)		lsyf	21	D T V K L M Y K G Q P M T F R L
2	12	0.59	1ttbA	33	F R K A A D D T W E P F
	(5)		lpyaB	109	T E T Q W D G S E L P I
			lalkA	389	A L N T K D G A V M V M
			2acq	3	R L L L N N G A K M P I
			2pia	46	T V A V P N G S R R T Y
3	20	1.08	8abp	41	P D G E K T L N A I D S L A A S G A K G
	(3)		8abp	204	M N D S T V L G G V R A T E G Q G F K A
			lpdf	78	R D E N I R A V A I E N L K K R G I D A
4	23	1.88	8abp	41	P D G E K T L N A I D S L A A S G A K G F V I
	(2)		lpdf	78	R D E N I R A V A I E N L K K R G I D A L V V
5	20	0.91	lbfq	27	N G G F F L R I H P D G R V D G V R E K
	(2)		lbfq	69	S A N R Y L A M K E D G R L L A S K S V
6	12	1.05	larb	197	P I Y S P E K R V L G Q
	(3)		lcdh	27	H W K N S N Q I K I L G
			lhbq	121	R L L N L D G T C A D S
7	15	0.91	2aaiB	179	G S I R P Q Q N R D N C L T S
	(2)		lbfq	86	N T Y R S R K Y T S W Y V A L
8	19	0.24	9wgaA	17	C C S Q Y G Y C G M G G D Y C G K G C
	(4)		9wgaA	60	C C S Q Y G H C G F G A E Y C G A G C
			9wgaA	103	C C S Q W G F C G L G S E F C G G G C
			9wgaA	146	C C S K W G S C G I G P G Y C G A G C
9	12	0.82	lmctA	199	V V C N G Q L Q G I V S
	(2)		lomp	97	V R Y N G K L I A Y P I
10	12	0.51	2hipA	29	Q L C E N C A F W G E A
	(2)		lisuA	20	K K C S G C A Q F V P G
11	12	0.72	ladd	265	S S Y L T G A W D P K T
	(2)		lomp	197	D L I K N K H M N A D T
12	12	0.54	2gbp	131	Q G W D L N K D G Q I Q
	(3)		2scpA	101	R A V D T N E D N N I S
			2scpA	135	D A I D T N N D G L L S

The table gives for each pattern its numeration, its length (L), together with the number of repetitions in the database (Nb) and the associated average pairwise C_{α} <r.m.s.> deviation. The right-hand side of the table provides for each (1) its protein PDB code, followed by the position of the first residue in the PDB entry, and (2) the full amino acid sequence with conserved patterns indicated in grey.

1993; Schuchhardt *et al.*, 1996; Bystroff and Baker, 1998) have described larger fragments (6–20 residues) and searched for an ‘exhaustive’ description of local structures. They have then to deal with about 100 classes. Among studies searching for a limited number of 3D local structures, whereas others worked on larger fragments, only Rooman *et al.* (1990) used short fragments of four residues and chose a rough classification of four classes to study sequence patterns associated with the classes. In this study, we have focused on a limited number of blocks obtained from short fragments of four residues each, to assess of what interest may be the integration of mechanisms underlying the connections between representative SBBs.

The HMM seems to capture general features of protein structures, such as the observed length distribution of helices and strands. The most rigid and dimensionally well-defined classical secondary structure α -helix (Richardson and Richardson, 1988) was well identified by two SBBs: one block appears to correspond to the core of α -helices. The other does not correspond to standard well-known π or 3.10 helices but rather to more irregular and stretched helices. β -Strands appear to fall into two categories, reflecting the remarkable range of average length and structural variability among β -strands (Orengo *et al.*, 1994). This is in agreement with previous studies (Rooman *et al.*, 1990; Prestrelski *et al.*, 1992; Schuchhardt *et al.*, 1996). Concerning flanking regions, Fetrow *et al.* (1997)

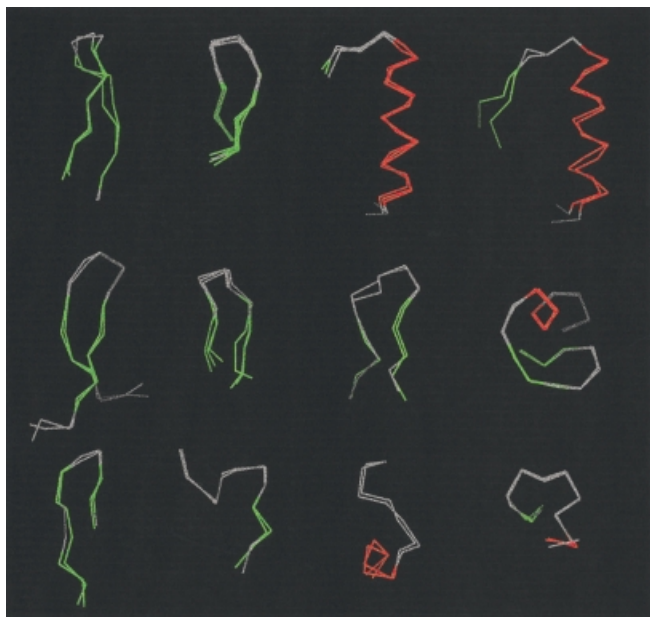


Fig. 5. Illustration of 12 extracted long structural blocks (LSBs), i.e. contiguous backbone chain fragments of defined length, that formed words made up of identical SBBs in the database of non-homologous proteins. These LSBs include more than 11 SBBs ($L \geq 11$) and are repeated more than twice ($Nb \geq 2$). The LSBs are positioned according to their numbering in Table III, from left to right, top to bottom. They are coloured according to their corresponding secondary structure. The figure was generated using Xmol (Tufféry, 1995).

demonstrated that both helix and strand capping structures can be objectively recognized by their local C_α geometry; the HMM also seems able to classify and subclassify distinct structural categories for the blocks flanking both α -helices and β -strands. More generally, on increasing the number of states up to 12, the HMM progressively classifies α -helices into two more or less regular structures, differentiates β -strands into two structures and decomposes coils.

Interest and limits of the HMM

In contrast to supervised learning strategies (Levitt and Chothia, 1976; Kabsch and Sander, 1983; Richards and Kundrot, 1988; Prestrelski *et al.*, 1992; Hutchinson and Thornton, 1993; Zhu, 1995), the structural blocks emerged from the HMM without any *a priori* knowledge of secondary structural classification. In that sense, the HMM is able to classify conformations that template studies must describe as undefined or random structures and also to subdivide conformation classes previously defined as a single class. Moreover, all segments are assigned to one SBB by this approach: there are no unclassified segments. Compared with the usual clustering approaches (Rooman *et al.*, 1989, 1990, 1991; Unger *et al.*, 1989; Fechter *et al.*, 1995; Pavone *et al.*, 1996; Bystroff and Baker, 1998), the HMM approach enables us to distinguish separate states, without making *a priori* choices (Kelley *et al.*, 1996). For instance, the HMM approach allows different levels of variability within each SBB: some identified SBBs are well defined and have little variability (r.m.s.d.w. < 0.35 Å) and clear orientation whereas others are fuzzier (r.m.s.d.w. < 1 Å), with no distinct orientation. Learning methods based on networks (Schuchhardt *et al.*, 1996; Fetrow *et al.*, 1997) have recently been used to extract and classify local protein backbone elements but these methods did not take into account any local dependence between blocks: all these studies used only the

structural characteristics to identify structural 3D blocks and reconstructed *a posteriori* the organization of these 3D conformations. One major contribution of HMM is that this model implicitly takes into account the sequential connections between the SBBs. It is striking that structurally close SBBs can have different roles in the construction of building blocks.

Throughout this study, the underlying Markov chain was set as first order. It would be interesting to consider Markov chains of higher order, i.e. the dependence between more than two consecutive SBBs might be informative. The increased size of the transition matrix and its interpretation will raise problems, but this could be achieved for a reasonable number of structural blocks.

Concerning the 'coding' of the proteins, the HMM directly estimates the most probable series of SBBs taking into account the underlying mechanism in these blocks (see Materials and methods), whereas other approaches have to deal with multiple ways to combine correctly the identified short fragments, step by step, into a full protein (Nakashima *et al.*, 1988; Richards and Kundrot, 1988; Unger *et al.*, 1989; Schuchhardt *et al.*, 1996; Fetrow *et al.*, 1997). Although the local interaction between consecutive fragments is not large enough to dictate the global conformation of the entire protein, it is an interesting step to be able to reconstruct proteins as series of SBBs taking into account sequential connectivities of these 3D conformations.

Finally, some limitations of the approach are inherent in its practical implementation. We have assumed, based on biological variability considerations, that each protein has its own initial block, (i.e. there is an initial law of the Markov chain specific to each protein) and not the existence of a unique initial block common for all proteins. This results in a drastic increase in the number of parameters when the number of states (a function of the product of the number of proteins times the number of states minus one) is increased and limits the number of proteins used to identify the SBB code. However, the degree to which the algorithm employed here is dependent on the number of proteins in the dataset studied was examined in applying the HMM approach to two independent libraries of 50 and one of 100 proteins each (see Results).

Conclusion and perspectives

We have presented a stochastic HMM approach to the problems of (i) characterizing different short structural 3D building blocks, (ii) describing the heterogeneity of their corresponding short segments and (iii) studying their global organization by quantifying their connections. The transition matrix shows only a limited number of transitions between SBBs, indicating that the succession of SBBs is not stochastic and that the connections by which the blocks form the protein structure are well organized. Structural and sequential observations together show that HMM is able to identify distinct SBBs, corresponding to regular secondary structures, their flanking regions or coil regions. Isolated SBBs exhibit different sequence specificity, even if only some have a strong signal. The sequences associated with the SBBs are overall compatible with amino acid preferences already described for the regular secondary structures and their capping regions. Finally, preliminary observations suggest that SBBs can make sense in terms of the structural alphabet. All these results indicate that the HMM is a promising tool for decoding the framework of the protein backbone. Combining sequence specificity and

transitions learned by HMM could be promising with respect to structure prediction.

Work is in progress to exploit the SBB code on a larger database of proteins: first, to analyse more accurately sequence specificity associated with SBBs, and second, to extract an exhaustive catalogue of similar fragments based on repetitive series of SBBs.

References

- Argos, P. and Palau, J. (1982) *Int. J. Pept. Protein Res.*, **19**, 380–393.
- Asai, K., Hazamizu, S. and Handa, K. (1993) *CABIOS*, **9**, 141–146.
- Aurora, R., Srinivasan, R. and Rose, G.D. (1994) *Science*, **264**, 1126–1130.
- Baum, L.E., Petrie, T., Soules, G. and Weiss, N. (1970) *Ann. Math. Stat.*, **41**, 164–171.
- Berstein, F.C., Koetzle, T.G., Williams, G.J.B., Meyer, E.F., Brice, M.D., Rogers, J.R., Kennard, O., Shimanouchi, T. and Tasumi, M. (1977) *J. Mol. Biol.*, **112**, 535–542.
- Bystroff, C., Baker, D. (1998) *J. Mol. Biol.*, **281**, 356–377.
- Camproux, A.C., Saunier, F., Chouvet, G., Thalabard, J.C. and Thomas, G. (1996) *Biophys. J.*, **71**, 2404–2412.
- Camproux, A.C., Tufféry, P., Buffat, L., André, C., Boisvieux, J.F. and Hazout, S. (1999) *Theor. Chem. Acc.*, **101**, 33–40.
- Churchill, G.A. (1989) *Bull. Math. Biol.*, **51**, 79–94.
- Colloc'h, N., Etchebest, C., Thoreau, E., Henrissat, B. and Mornon, J. (1993) *Protein Engng*, **6**, 377–382.
- Di Francesco, V., Garnier, J. and Munson, P.J. (1997) *J. Mol. Biol.*, **267**, 446–463.
- Efimov, A.V. (1984) *FEBS Lett.*, **166**, 33–38.
- Fechteler, T., Dengler, U. and Schomburg, D. (1995) *J. Mol. Biol.*, **253**, 114–131.
- Fetrow, J.S. (1995) *FASEB J.*, **9**, 708–717.
- Fetrow, J.S., Palumbo, M.J. and Berg, G. (1997) *Proteins*, **27**, 249–271.
- Harper, E.T. and Rose, G.D. (1993) *Biochemistry*, **32**, 7605–7609.
- Hobohm, U., Scharf, M., Schneider, M. and Sandres, C. (1992) *Protein Sci.*, **1**, 409–417.
- Hutchinson, E.G. and Thornton, J.M. (1993) *Protein Engng*, **6**, 233–245.
- Kabsch, W. and Sander, C. (1983) *Biopolymers*, **22**, 2577–2637.
- Karplus, K., Sjölander, K., Barrett, C., Cline, M., Haussler, D., Hughey, R., Holm, L. and Sander, C. (1997) *Proteins*, Suppl 1, 134–139.
- Kelley, L.A., Gardner, S.P. and Sutcliffe, M.J. (1996) *Protein Engng*, **9**, 1063–1065.
- Krogh, A., Brown, M., Mian, S., Sjölander, K. and Haussler, D. (1994) *J. Mol. Biol.*, **235**, 1501–1531.
- Leszczynski, J.F. and Rose, G.D. (1986) *Science*, **234**, 849–855.
- Levitt, M. and Chothia, C. (1976) *Nature*, **261**, 552–542.
- Nakashima, H., Nishikawa, K. and Ooi, T. (1988) *J. Protein Chem.*, **7**, 509–525.
- Orengo, C.A., Flores, T.P., Taylor, W.R. and Thornton, J.M. (1993) *Protein Engng*, **6**, 485–500.
- Orengo, C.A., Jones, D.T. and Thornton, J.M. (1994) *Nature*, **372**, 631–635.
- Pavone, V., Gaeta, G., Lombardi, A., Nastri, F., Maglio, O., Isernia, C. and Saviano, M. (1996) *Biopolymers*, **38**, 705–721.
- Presta, L. (1989) *Protein Engng*, **2**, 395–397.
- Prestrelski, S.J., Williams, A.L. and Liebman, M.N. (1992) *Proteins*, **14**, 430–439.
- Rabiner, L.R. (1989) *Proc. IEEE*, **77**, 257–285.
- Rackovsky, S. (1993) *Proc. Natl Acad. Sci. USA*, **90**, 644–648.
- Richards, F.M. and Kundrot, C.E. (1988) *Proteins*, **3**, 71–84.
- Richardson, J.S. and Richardson, D.C. (1988) *Science*, **240**, 1648–1652.
- Rooman, M.J., Wodak, S.J. and Thornton, J.M. (1989) *Protein Engng*, **3**, 23–27.
- Rooman, M.J., Rodriguez, J. and Wodak, S.J. (1990) *J. Mol. Biol.*, **213**, 327–336.
- Rooman, M.J., Kocher, J.-P.A. and Wodak, S.J. (1991) *J. Mol. Biol.*, **221**, 961–979.
- Schuchhardt, J., Schneider, G., Reichelt, J., Schomburg, D. and Wrede, P. (1996) *Protein Engng*, **9**, 833–842.
- Schwartz, G. (1978) *Ann. Stat.*, **6**, 461–464.
- Seales, J.W., Srinivasan, R. and Rose, G.D. (1994) *Protein Sci.*, **3**, 1741–1745.
- Smith, P.E., Blatt, H.D. and Pettitt, B.M. (1997) *Proteins*, **27**, 227–234.
- Sonnhammer, E.L., Eddy, S.R., Birney, E., Bateman, A. and Durbin, R. (1998) *Nucleic Acids Res.*, **26**, 320–322.
- Stultz, C.M., White, J.V. and Smith, T.F. (1993) *Protein Sci.*, **2**, 305–314.
- Tufféry, P. (1995) *J. Mol. Graphics*, **13**, 67–72.
- Unger, R., Harel, D., Wherland, S. and Sussman, J.L. (1989) *Proteins*, **5**, 355–373.
- Unger, R. and Sussman, J.L. (1993) *J. Comput.-Aided Mol. Des.*, **7**, 457–472.
- Zhu, Z. (1995) *Protein Engng*, **8**, 103–108.

Received December 23, 1998; revised August 19, 1999; accepted September 1, 1999